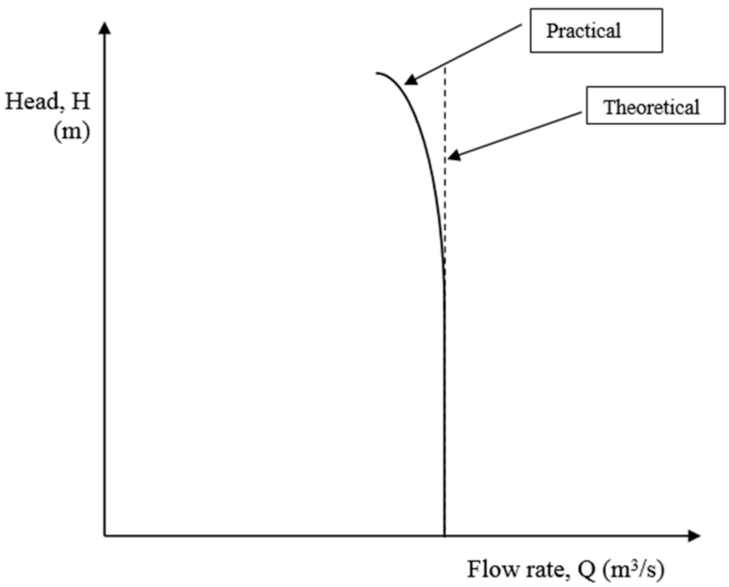
Positive Displacement Pumps

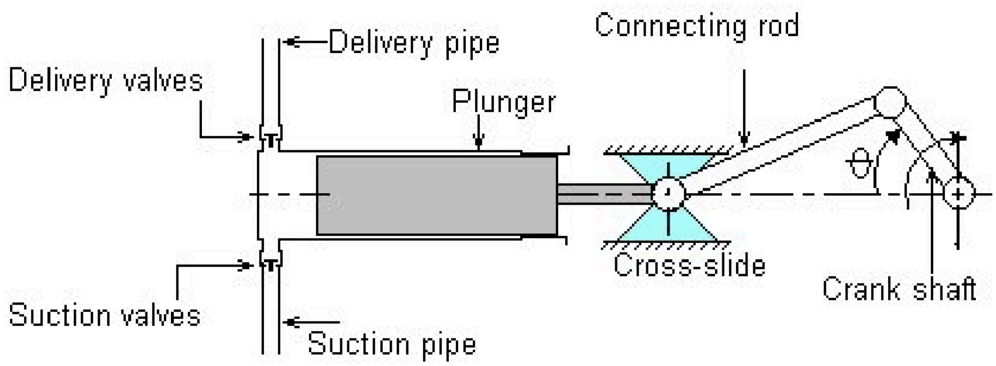
A positive displacement pump is a pump that delivers a fixed amount of fluid per revolution of the transmission shaft. Typically, there are two types of positive displacement pumps, reciprocating pumps and rotary pumps.

As a fixed amount of fluid is supplied per revolution, the volumetric flow rate remains constant, even for variable pressure (or variable head). Thus, on the Q-H graph, the curve is a vertical straight line. However, in practical cases, higher pressures causes leakages past seals, causing a decrease in volumetric flow rate.



The flow rate value of the pump depends on the speed of the transmission shaft, being proportional to the speed.

A typical single acting reciprocating positive displacement pump is shown below:



On the plunger's reverse stroke, it creates a pressure drop in the chamber, which opens the suction valve and closes the delivery valve, allowing for a fixed amount of fluid to be sucked in, before being pumped out on the forward stroke of the plunger, while opening the delivery valve and closing the suction valve.

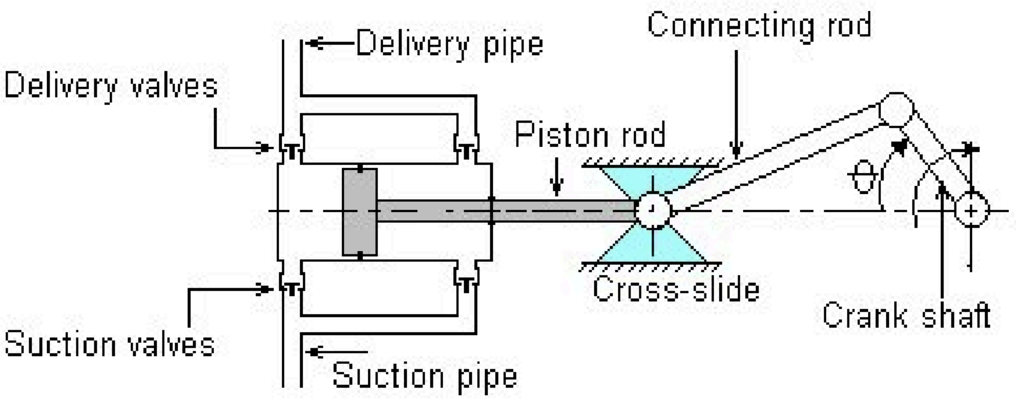
The volume of fluid the pump displaces out is the displaced volume of the plunger, which is

$$V = AL = \frac{\pi}{4} D^2 S$$

As this represents the amount of volume displaced in 1 revolution of the crank shaft, the volumetric flow rate can be found as

$$Q = V[m^3/rev] \times \omega[rev/s]$$
$$Q = V \times \frac{N}{60}$$
$$= \frac{\pi}{4} \times D^2 \times S \times \frac{N}{60}$$

A double acting reciprocating pump works by using the reverse stroke of pump to displace fluid.



Thus, the fluid displaced in one revolution is due to both the front and back of the piston, which is

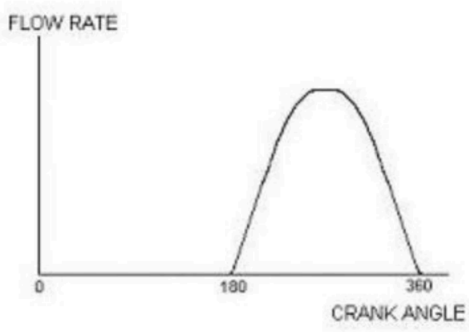
$$V = \left(\frac{\pi}{4}D^2 + \frac{\pi}{4}(D^2 - d^2)\right) S$$

where d is the diameter of the piston rod.

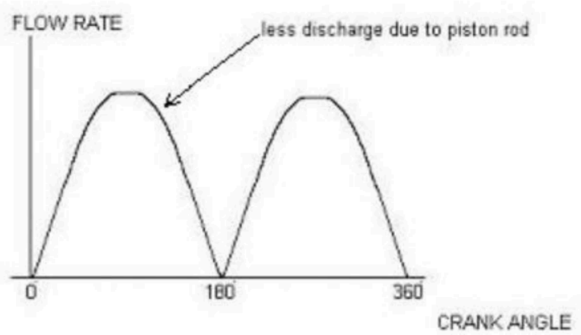
Thus, the volumetric flow rate is

$$\begin{aligned} Q &= V \times \frac{N}{60} \\ &= \left(\frac{\pi}{4}D^2 + \frac{\pi}{4}(D^2 - d^2)\right) S \times \frac{N}{60} \end{aligned}$$

As the shaft rotates, the piston does not move at a constant velocity, but instead varies by the sine of the angle of the crankshaft, thus the flow rate will not be constant but pulsate.



Flow vs crank angle for a single acting pump

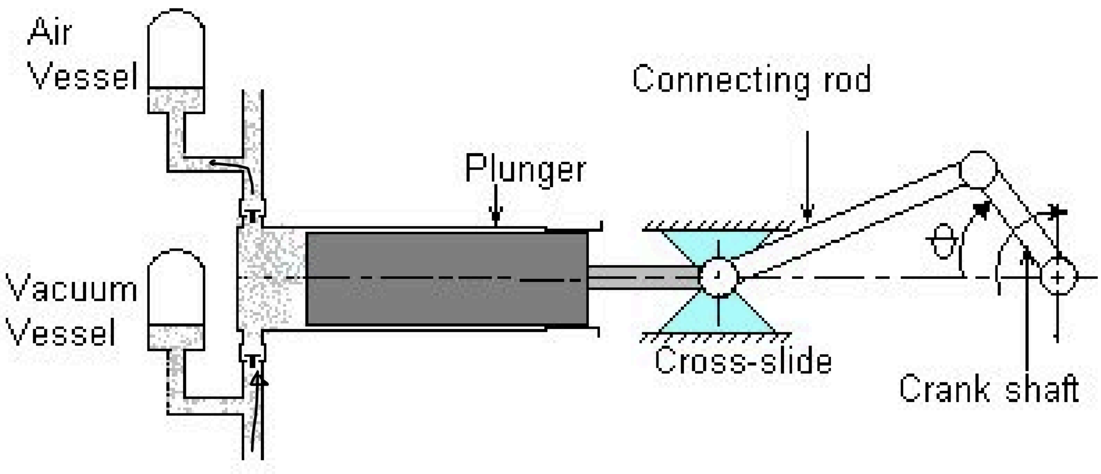


Flow vs crank angle for a double acting pump

For double acting pumps, the peak of the first pulse will be slightly lesser than the second pulse, due to the lesser area of the piston from the piston rod.

As this may not be desirable is some applications, two ways of reducing the pulsation are by using air/vacuum vessels, and using multi-cylinder pumps.

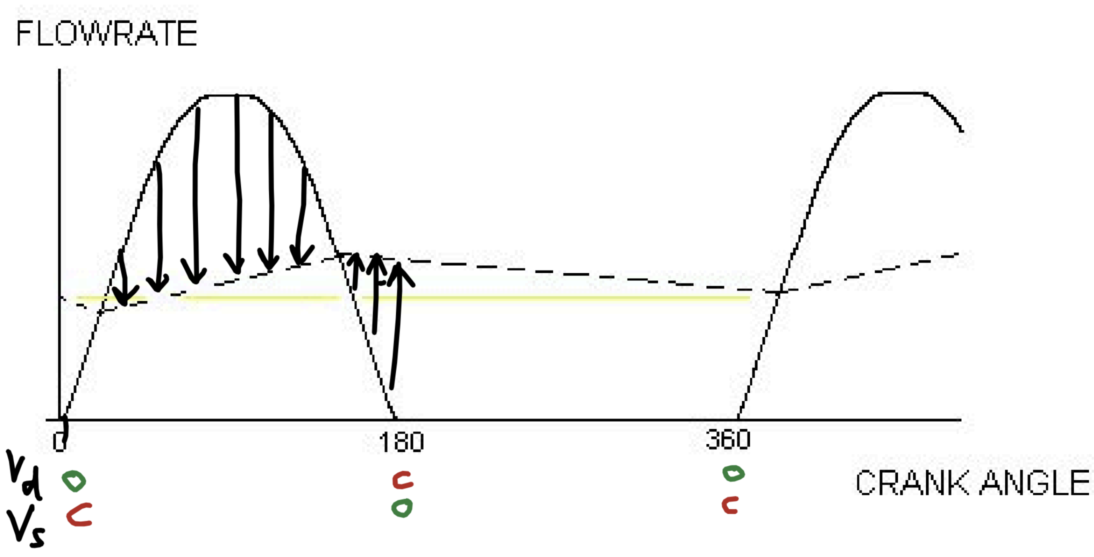
The air/vacuum vessels method involves adding a vacuum vessel at the upstream (before the pump) and an air vessel at the downstream (after the pump).



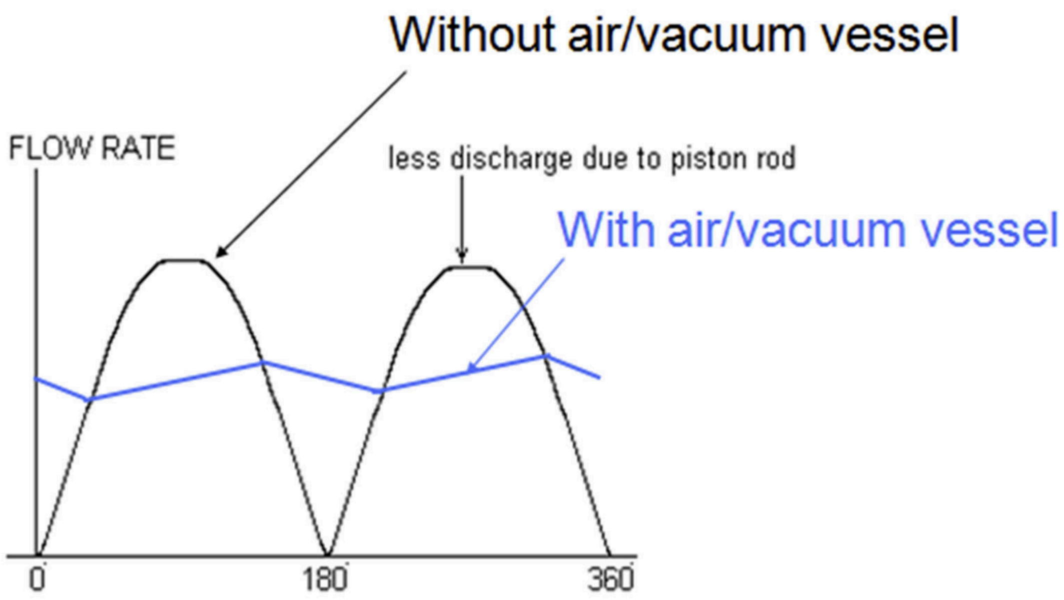
For the air vessel, it helps to regulate the volumetric flow when both the delivery valve is open or not. When the delivery valve is open, it allows some fluid to flow into the air vessel to reduce the flow rate when the flow rate becomes higher. When the valve is closed, the air vessel has a higher fluid level, and thus is able to supply some fluid due to the pressure of the air vessel pushing out the fluid.

For the vacuum vessel, when the suction valve is open, the suction of the pump suctions up fluid from both the vessel and the pipe, increasing the flow rate. When the suction valve closes, the flow rate from the pipe is still constant, but instead is directed into the vacuum vessel (which is able to hold fluid due to the vacuum)

Graphically, on the $Q - \theta$ graph, the flowrate oscillates less, remaining at a relatively constant amount. What should be noted is that both air and vacuum vessels work in tandem, as these valves are not just at fixed open or closed states, but continuously open and close, allowing each vessel to operate their function.

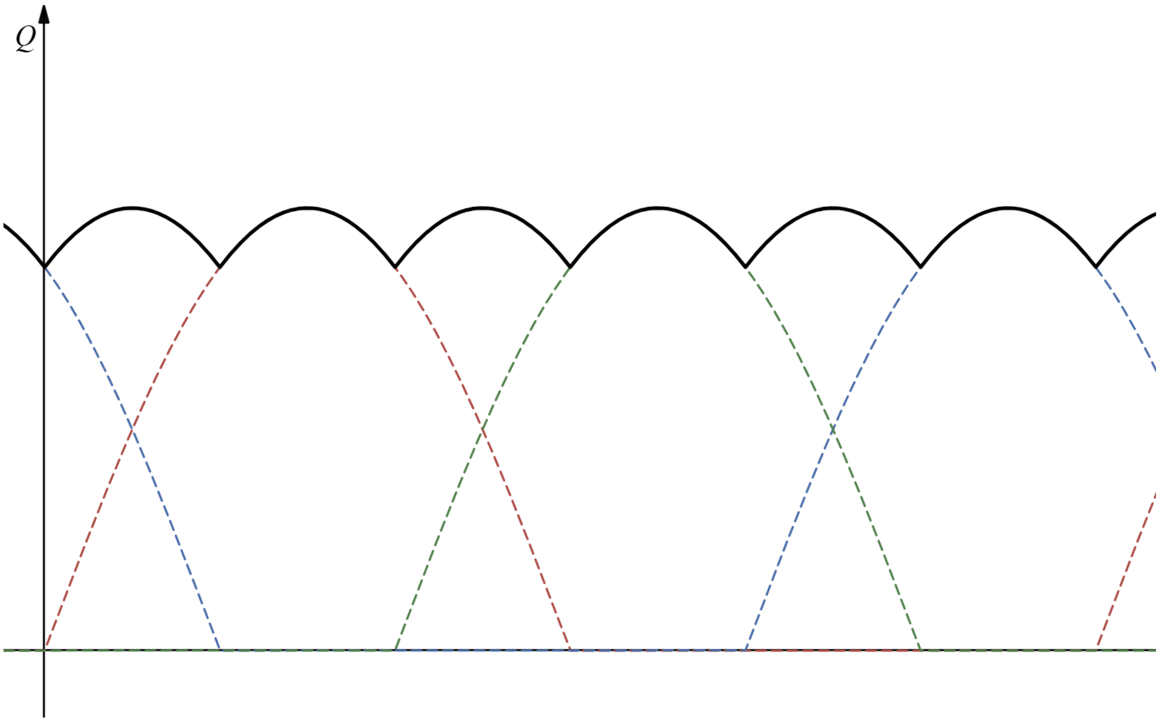


A double acting pump has a similar process, where it is able to supply flow rate even when the delivery valve is closed due to the air vessel.



Another method is by using multi-cylinder pumps. As a single cylinder can only produce one phase of flow rates, if we set up more cylinders at a different phase offset, the sum of their flowrates would be constant (or roughly constant) over time.

For example, shown is three pistons connected to the same crank shaft, with a 120 degree phase offset, known as a triplex arrangement.



The flowrate is now more consistent than one cylinder, oscillating at a lesser amplitude but more frequently. Other setups also include more pistons, like 4 (quadruplex) pistons.